

```
"""
```

```
grus_constants.py
```

```
=====
```

```
Green Recursive Utility Service LLC – Audit Engine Constants
```

```
Specification: GRUS-CRX-001-v1.0
```

```
Public charter: GRUS-01
```

```
This module is the single source of truth for the GRUS audit pipeline  
configuration. The audit engine imports from this module only; nothing  
is hardcoded outside it.
```

```
Published by: Green Recursive Utility Service LLC
```

```
State of Texas LLC | Filed May 5, 2026
```

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"""
```

```
# ——— LLC IDENTITY —————
```

```
LLC_NAME = "Green Recursive Utility Service LLC"
```

```
LLC_STATE = "Texas"
```

```
LLC_FILING_DATE = "2026-05-05"
```

```
LLC_PRINCIPAL_OFFICE = "Weatherford, Texas, USA"
```

```
# ——— AUDIT PIPELINE VERSION —————
```

```
AUDIT_PIPELINE_VERSION = "1.0"
```

```
CERTIFICATION_STANDARD = "GRUS-STD-001-v1.0"
```

```
ENGINE_REPOSITORY = "github.com/grus-utility/grus-audit-engine"
```

```
# ——— CRYPTOGRAPHIC KEY LOCATIONS —————
```

```
GRUS_PUBLIC_KEY_CANONICAL = "https://grus.utility/public-key"
```

```
GRUS_PUBLIC_KEY_MIRRORS = [
```

```
    "https://github.com/grus-utility/public-key",
```

```
    "https://zenodo.org/communities/grus-utility",
```

```
    "https://web.archive.org/web/grus.utility/public-key",
```

```
]
```

```
# ——— PUBLIC LEDGER LOCATIONS —————
```

```
LEDGER_CANONICAL_URL = "https://ledger.grus.utility"
```

```
LEDGER_GITHUB_MIRROR = "https://github.com/grus-utility/grus-ledger"
```

```
LEDGER_ZENODO_COMMUNITY = "grus-ledger"
```

```
LEDGER_INTERNET_ARCHIVE = "https://web.archive.org/web/ledger.grus.utility"
```

```
# ——— SVCF FOUNDATION REFERENCES (PRESERVED, NOT MODIFIED) —————
```

```
# These are public-archive repositories. The LLC validates and certifies them  
# but does not modify, relocate, or rebrand them.
```

```
SVCF_FOUNDATION_REPO = "<inaugural_certification_target_repository_url>"
```

```
SVCF_PRIORITY_ANCHOR = "rxiVerse:2602.0018"
```

```
SVCF_PRIORITY_DATE = "2025-11-16"
```

```
SVCF_FOUNDATION_DOIS = [
```

```
    "10.5281/zenodo.18604376",
```

```

"10.5281/zenodo.18848748",
"10.5281/zenodo.198940",
]
SVCF_CODE_RELEASE = "rxiv:2604.0118"
SVCF_CODE_RELEASE_DATE = "2026-04-29"
SVCF_FOUNDATIONAL_AUDIT_ID = "GRUS-AUDIT-0001"

# — EIGHT LOCKED CONSTANTS (SVCF FOUNDATION – IMMUTABLE) —
# Imported reference only. The audit pipeline reads these from the
# submission's svcf_constants.py and verifies they match these exact values.
EXPECTED_CONSTANTS = {
    "GAMMA": {"value": "1.0/2857.0", "type": "MEASURED",
              "source": "STAR  $\Lambda$ - $\bar{\Lambda}$  vorticity, Au+Au  $\sqrt{s_{NN}}$ =200 GeV, Nature 2026"},
    "B": {"value": "32.0/33.0", "type": "EXACT",
          "source": "BRST:  $B = (N-5)/(N-4)$  at  $N=37$ "},
    "BETA": {"value": "65.0/66.0", "type": "EXACT",
             "source": "Law #1:  $\beta = (B+1)/2$ "},
    "PSI": {"value": "sqrt(2.0) - 1.0", "type": "EXACT",
            "source": "37D  $\rightarrow$  4D  $S^2$  projection integral"},
    "K_TD": {"value": "37 * 300", "type": "EXACT",
             "source": "37  $\times$  C(25,2) = 11100"},
    "K_MODE": {"value": "9", "type": "EXACT",
               "source": "St-Jean photonic Chern insulator (PRX 2026)"},
    "RHO_C": {"value": "1.01e-26", "type": "MEASURED",
              "source": " $H_0=70$  km/s/Mpc, Friedmann critical-density relation"},
    "ETA": {"value": "6.8e-28", "type": "MEASURED",
            "source": "CHIME FRB pulse-broadening slope"},
    "EPSILON": {"value": "ALPHA**2 = 5.3251e-5", "type": "DERIVED",
                "source": "Law #2: double-vertex EM coupling"},
}

# — AUDIT PIPELINE STAGES —
PIPELINE_STAGES = [
    "manifest_parsing",
    "constants_validation",
    "doi_resolution",
    "dataset_fetching",
    "verify_execution",
    "residual_comparison",
    "closure_testing",
    "falsification_validation",
    "hashing",
    "certification_issuance",
]

# — DATA INTEGRITY CLAUSES —
INTEGRITY_CLAUSES = {
    "clause_1_zero_free_parameters": "All constants either EXACT or MEASURED with cited source",
    "clause_2_zero_placeholders": "No undeclared or 'to be determined' values",
    "clause_3_public_data_reproducibility": "Verification reproduces against public datasets",
}

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"clause_4_overconstained_closure": "Same constants close all touched domains",
"clause_5_forward_facing_falsification": "Predictions have explicit falsification
conditions",
"clause_6_immutable_public_timestamping": "Submission deposited at public archive with DOI",
}

# ——— RESIDUAL TOLERANCE ———
TOLERANCE_RELATIVE_PCT = 1.0 # residuals must match to within 1% relative
TOLERANCE_ABSOLUTE_SIGMA = 0.05 # or 0.05σ absolute, whichever is more permissive

# ——— COMPLIANCE FRAMING ———
COMPLIANCE_MANDATES = {
    "EO_14303": "Restoring Gold Standard Science (May 23, 2025)",
    "OSTP_NELSON_MEMO": "Free, Immediate, and Equitable Access to Federally Funded Research
(2022/2026)",
    "DATA_INTEGRITY_ACT": "Deterministic gradient validation, zero-proxy execution",
}

def print_config():
    """Print the LLC audit pipeline configuration."""
    print("=" * 70)
    print(f" {LLC_NAME}")
    print(f" Audit Pipeline Configuration v{AUDIT_PIPELINE_VERSION}")
    print("=" * 70)
    print()
    print(f" SVCF Foundation (preserved as filed):")
    print(f"   Author: {SVCF_AUTHOR}")
    print(f"   Repository: {SVCF_FOUNDATION_REPO}")
    print(f"   Priority anchor: {SVCF_PRIORITY_ANCHOR} ({SVCF_PRIORITY_DATE})")
    print(f"   Code release: {SVCF_CODE_RELEASE} ({SVCF_CODE_RELEASE_DATE})")
    print()
    print(f" LLC Audit Engine:")
    print(f"   Repository: {ENGINE_REPOSITORY}")
    print(f"   Standard: {CERTIFICATION_STANDARD}")
    print(f"   Pipeline stages: {len(PIPELINE_STAGES)}")
    print(f"   Integrity clauses: {len(INTEGRITY_CLAUSES)}")
    print()
    print(f" Compliance:")
    for k, v in COMPLIANCE_MANDATES.items():
        print(f"   {k}: {v}")
    print()
    print(f" Public ledger: {LEDGER_CANONICAL_URL}")
    print(f" Public key: {GRUS_PUBLIC_KEY_CANONICAL}")
    print()

if __name__ == "__main__":
    print_config()

```

